Maven ⭐ ⭐ ⭐ — Interview-Ready Q&A

1. What is Maven?
Maven is a build automation and dependency management tool mainly used for Java applications.
It helps developers compile source code, run tests, package applications, manage dependencies, and create deployable artifacts.
In DevOps, Maven is commonly integrated with Jenkins to automate Java application builds.
Simple ga gurthupettuko:
Maven = Java Build + Dependency Management Tool
Eng: Maven is a build automation and dependency management tool commonly used for Java projects and CI/CD pipelines.

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2. What is pom.xml**?**
pom.xml stands for **Project Object Model** file.
It is the main configuration file of a Maven project and contains information such as project details, dependencies, plugins, build configuration, and packaging type.
Maven reads this file to understand how the project should be built.
Simple ga gurthupettuko:
pom.xml = Maven project ki main configuration file.
Eng: pom.xml is the main Maven configuration file that defines project information, dependencies, plugins, and build configuration.

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3. What are Maven Dependencies?
Dependencies are external libraries required by an application during compilation, testing, or runtime.
Instead of manually downloading JAR files, we define the required dependencies in pom.xml, and Maven downloads them automatically from configured repositories.
Simple ga gurthupettuko:
Dependency = Application ki required external library.
Eng: Maven dependencies are external libraries required by the application and managed through pom.xml.

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4. What is Maven Repository?
A Maven repository is a location where Maven stores and retrieves project artifacts and dependencies.
There are three common repository types: **Local Repository, Central Repository, and Remote/Private Repository**.
For example, organizations may use Nexus or Artifactory as a private Maven repository.
Simple ga gurthupettuko:
Repository = Dependencies and artifacts store ayye place.
Eng: A Maven repository stores and provides dependencies and build artifacts.

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5. What is Local Repository?
The local repository is a directory on the developer or build machine where Maven caches downloaded dependencies and artifacts.
By default, on Linux it is commonly located under ~/.m2/repository.
If a required dependency is already available locally, Maven can reuse it instead of downloading it again.
Simple ga gurthupettuko:
Local Repository = Mana machine lo Maven cache.
Eng: The Maven local repository stores downloaded dependencies and artifacts on the local machine.

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6. What is Maven Central Repository?
Maven Central is a widely used public repository containing many Java and JVM-related libraries and artifacts.
When a dependency is not available in the local repository, Maven can retrieve it from configured remote repositories such as Maven Central.
Simple ga gurthupettuko:
Maven Central = Public repository for commonly available dependencies.
Eng: Maven Central is a public repository that provides many commonly used Java dependencies and artifacts.

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7. What is a Private Maven Repository?
A private Maven repository is an organization's internal repository used to store proprietary dependencies and build artifacts.
Tools such as Nexus Repository and JFrog Artifactory can be used for this purpose.
It gives organizations better control over artifact storage, access, and distribution.
Simple ga gurthupettuko:
Private Repo = Company internal artifacts/dependencies store.
Eng: A private Maven repository stores and manages an organization's internal dependencies and build artifacts.

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8. What is Maven Lifecycle?
A Maven lifecycle is a predefined sequence of phases used to build and manage a project.
The commonly used lifecycle includes phases such as **validate, compile, test, package, verify, install, and deploy**.
When we execute a later phase, Maven normally executes the required earlier phases as well.
Simple ga gurthupettuko:
Lifecycle = Maven build process sequence.
Eng: Maven lifecycle defines the sequence of phases used to validate, compile, test, package, and deploy a project.

9. What is the difference between Maven Phase and Goal?
A **phase** is a step in the Maven lifecycle, such as compile, test, or package.
A **goal** is a specific task provided by a Maven plugin.
For example, the compiler:compile goal is executed as part of the compile phase.
Simple ga gurthupettuko:
Phase = Lifecycle step
Goal = Plugin task
Eng: A Maven phase is part of the lifecycle, while a goal is a specific task provided by a plugin.

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10. What is mvn clean**?**
mvn clean removes files generated by previous Maven builds, mainly the target directory.
It is useful when we want to perform a fresh build without using previous build output.
In CI/CD pipelines, clean builds can help avoid issues caused by stale build files.
Simple ga gurthupettuko:
mvn clean = Previous build output clean cheyyadam.
Eng: mvn clean removes previous Maven build output so the project can be built from a clean state.

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11. What is mvn compile**?**
mvn compile compiles the project's main source code.
Before compilation, Maven executes the required earlier lifecycle phases and resolves required dependencies.
The compiled classes are generally placed under the target directory.
Simple ga gurthupettuko:
compile = Source code → Compiled classes
Eng: mvn compile compiles the main source code of the Maven project.

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12. What is mvn test**?**
mvn test compiles the required code and executes the project's unit tests.
If the tests fail, Maven normally marks the build as failed.
In CI/CD, this helps ensure that code changes do not introduce known unit-test failures.
Simple ga gurthupettuko:
mvn test = Build code + Run unit tests.
Eng: mvn test runs the project's unit tests and fails the build when the tests fail.

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13. What is mvn package**?**
mvn package builds the application and packages it into the configured artifact format.
For example, a Java application can produce a .jar or .war file depending on the project configuration.
The generated artifact is normally placed inside the target directory.
Simple ga gurthupettuko:
package = Application ni deployable artifact ga create cheyyadam.
Eng: mvn package builds and packages the application into a JAR, WAR, or configured artifact format.

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14. What is mvn install**?**
mvn install builds and packages the project and then installs the generated artifact into the local Maven repository.
This allows other local Maven projects to use that artifact as a dependency.
Simple ga gurthupettuko:
install = Build → Package → Local Repository lo store.
Eng: mvn install builds the project and installs its artifact into the local Maven repository.

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15. What is mvn deploy**?**
mvn deploy builds and packages the project and publishes the generated artifact to a configured remote Maven repository.
This is commonly used in CI/CD pipelines to publish application artifacts to repositories such as Nexus or Artifactory.
Simple ga gurthupettuko:
deploy = Artifact ni Remote Repository ki upload cheyyadam.
Eng: mvn deploy publishes the generated Maven artifact to a configured remote repository.

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16. What is the difference between mvn install **and** mvn deploy**?**
mvn install stores the generated artifact in the local Maven repository.
mvn deploy publishes the artifact to a remote repository so that other systems or teams can consume it.
Simple ga gurthupettuko:
install = Local
deploy = Remote
Eng: Install puts the artifact into the local repository, while deploy publishes it to a remote repository.

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17. What are Maven Plugins?
Maven plugins provide the actual functionality used during the Maven build lifecycle.
They can compile code, run tests, package applications, generate reports, perform code analysis, and execute other build tasks.
For example, compiler and Surefire plugins are commonly used in Maven projects.
Simple ga gurthupettuko:
Plugin = Maven ki specific functionality provide chese component.
Eng: Maven plugins provide the tasks and functionality executed during the build lifecycle.

18. What is settings.xml **in Maven?**
settings.xml is used for Maven configuration that is generally external to the project itself.
It can contain repository configuration, mirrors, authentication-related configuration, proxies, and other Maven settings.
It can exist at the user level or Maven installation level.
Simple ga gurthupettuko:
pom.xml = Project configuration
settings.xml = Maven/user environment configuration
Eng: settings.xml contains Maven environment and repository-related configuration outside the project POM.

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19. What is the difference between pom.xml **and** settings.xml**?**
pom.xml contains project-specific configuration such as dependencies, plugins, packaging, and build settings.
settings.xml contains user or environment-specific Maven configuration such as mirrors, repositories, proxies, and credentials-related settings.
Simple ga gurthupettuko:
POM = Project specific
Settings = Environment/User specific
Eng: pom.xml defines project configuration, while settings.xml defines Maven user or environment configuration.

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20. What is Maven dependency scope?
Dependency scope determines where and when a dependency is available during the Maven build.
Common scopes include compile, provided, runtime, test, and system.
For example, a test dependency may be available only during testing and should not be included as a production runtime dependency.
Simple ga gurthupettuko:
Scope = Dependency ekkada/eppudu available undalo decide chestundi.
Eng: Dependency scope controls the classpath and lifecycle availability of a Maven dependency.

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21. What is transitive dependency?
A transitive dependency is a dependency that your project receives indirectly through another dependency.
For example, if library A depends on library B and your project depends on A, Maven can automatically resolve B as a transitive dependency.
This reduces the need to manually define every required library.
Simple ga gurthupettuko:
A → B → Mana Project ki B automatic ga ravachu.
Eng: A transitive dependency is an indirect dependency brought in through another project dependency.

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22. What is dependency conflict in Maven?
A dependency conflict occurs when different dependencies require different versions of the same library.
Maven uses dependency mediation rules to select a version, but the selected version may not always be what the application expects.
We should analyze the dependency tree and explicitly manage versions when necessary.
Simple ga gurthupettuko:
Same library → Multiple versions → Conflict possibility.
Eng: A dependency conflict occurs when different dependencies require different versions of the same library.

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23. How do you check Maven dependency tree?
We can use the Maven dependency tree functionality to view direct and transitive dependencies.
This helps identify which library introduced a dependency and can be very useful when troubleshooting version conflicts.
Simple ga gurthupettuko:
Dependency Tree = Which dependency ekkadi nunchi vachindo identify cheyyadam.
Eng: The Maven dependency tree helps identify direct and transitive dependencies and troubleshoot version conflicts.

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24. What is the target **directory in Maven?**
The target directory contains files generated during the Maven build.
It can contain compiled classes, test reports, packaged JAR/WAR files, and other build output.
The mvn clean command normally removes this directory.
Simple ga gurthupettuko:
target = Maven build output folder.
Eng: The target directory contains files generated during Maven builds, including compiled code and packaged artifacts.

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25. How do you skip tests in Maven?
Maven provides options to skip test execution when required.
This can be useful in specific build situations, although skipping tests should be done carefully because it removes an important validation step from the pipeline.
Simple ga gurthupettuko:
Skip tests = Fast build, but validation miss avvachu.
Eng: Maven supports options to skip test execution, but this should be used carefully because tests are an important CI validation step.

26. How do you integrate Maven with Jenkins?
Jenkins can check out Java source code from Git and execute Maven commands as part of the pipeline.
A typical pipeline can run:
Checkout → mvn clean → mvn test → mvn package → Quality Scan → Artifact Upload → Deploy
The generated artifact can then be stored in Nexus or another artifact repository.
Simple ga gurthupettuko:
Git → Jenkins → Maven Build → Test → Artifact → Deploy
Eng: Jenkins integrates with Maven by executing Maven build and test commands and publishing the generated artifacts as part of the CI/CD pipeline.

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27. How do you troubleshoot a Maven build failure?
First, I check the Jenkins or Maven console output and identify the exact failed phase or plugin.
Then I determine whether the issue is related to source code, dependency resolution, Java version, plugin configuration, repository connectivity, or environment configuration.
After fixing the root cause, I rerun the build and verify the generated artifact.
Simple ga gurthupettuko:
Failed Phase → Error Logs → Root Cause → Fix → Rebuild
Eng: I troubleshoot Maven failures by identifying the failed phase, analyzing the logs, checking dependencies and configuration, fixing the root cause, and rebuilding.

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28. What is Nexus Repository and how does Maven use it?
Nexus Repository is an artifact repository manager that can store and distribute build artifacts and dependencies.
Maven can download dependencies from Nexus and upload generated artifacts to it.
This allows an organization to maintain a centralized and controlled repository for its internal software components.
Simple ga gurthupettuko:
Maven ↔ Nexus ↔ Artifacts/Dependencies
Eng: Nexus acts as a centralized artifact repository that Maven can use to download dependencies and publish build artifacts.

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29. What is the difference between Snapshot and Release in Maven?
A **Snapshot** represents a development version that can change as development continues.
A **Release** represents a fixed version intended to be stable and should not normally be overwritten.
For example, 1.0-SNAPSHOT is a development version, while 1.0 can represent a released version.
Simple ga gurthupettuko:
SNAPSHOT = Changing development version
RELEASE = Stable fixed version
Eng: Snapshot versions are mutable development versions, while release versions are intended to be stable and immutable.

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30. Explain a Maven build flow in a real DevOps pipeline.
A developer pushes Java code to GitHub, which triggers Jenkins.
Jenkins checks out the code and runs Maven to clean, compile, test, and package the application. Code-quality or security checks can then be performed.
The generated JAR/WAR can be published to Nexus, and the approved artifact can be deployed to the target environment.
Simple ga gurthupettuko:
Developer → GitHub → Jenkins → Maven → Test → Package → Nexus → Deploy
Eng: In a typical DevOps flow, Jenkins uses Maven to build and test the Java application, publishes the artifact to a repository, and then deploys the approved artifact to the target environment.